Problem 3.9.4. Let A be an algebra, and let V be a representation of A. Let $\rho : A \to \mathrm{End}V$. A formal deformation of V is a formal series

$$\tilde{\rho} = \rho_0 + t\rho_1 + \cdots + t^n\rho_n + \ldots,$$

where $\rho_i : A \to \mathrm{End}(V)$ are linear maps, $\rho_0 = \rho$, and $\tilde{\rho}(ab) = \tilde{\rho}(a)\tilde{\rho}(b)$.

If $b(t) = 1 + b_1 t + b_2 t^2 + \ldots$, where $b_i \in \mathrm{End}(V)$, and $\tilde{\rho}$ is a formal deformation of ρ, then $b\tilde{\rho}b^{-1}$ is also a deformation of ρ, which is said to be isomorphic to $\tilde{\rho}$.

(a) Show that if $\mathrm{Ext}^1(V, V) = 0$, then any deformation of ρ is trivial, i.e., isomorphic to ρ.

(b) Is the converse to (a) true? (Consider the algebra of dual numbers $A = k[x]/x^2$.)

Problem 3.9.5. The Clifford algebra. Let V be a finite dimensional complex vector space equipped with a symmetric bilinear form $(\ ,\ )$. The **Clifford algebra** $\mathrm{Cl}(V)$ is the quotient of the tensor algebra TV by the ideal generated by the elements $v \otimes v - (v, v)1$, $v \in V$. More explicitly, if $x_i, 1 \le i \le N$, is a basis of V and $(x_i, x_j) = a_{ij}$ then $\mathrm{Cl}(V)$ is generated by x_i with defining relations

$$x_i x_j + x_j x_i = 2a_{ij},\ x_i^2 = a_{ii}.$$

Thus, if $(\ ,\ ) = 0$, $\mathrm{Cl}(V) = \wedge V$.

(i) Show that if $(\ ,\ )$ is nondegenerate, then $\mathrm{Cl}(V)$ is semisimple and has one irreducible representation of dimension 2^n if $\dim V = 2n$ (so in this case $\mathrm{Cl}(V)$ is a matrix algebra) and two such representations if $\dim(V) = 2n + 1$ (i.e., in this case $\mathrm{Cl}(V)$ is a direct sum of two matrix algebras).

Hint: In the even case, pick a basis $a_1, \ldots, a_n, b_1, \ldots, b_n$ of V in which $(a_i, a_j) = (b_i, b_j) = 0$, $(a_i, b_j) = \delta_{ij}/2$, and construct a representation of $\mathrm{Cl}(V)$ on $S := \wedge(a_1, \ldots, a_n)$ in which b_i acts as "differentiation" with respect to a_i. Show that S is irreducible. In the odd case the situation is similar, except there should be an additional basis vector c such that $(c, a_i) = (c, b_i) = 0$, $(c, c) = 1$ and the action of c on S may be defined either by $(-1)^{\mathrm{degree}}$ or by $(-1)^{\mathrm{degree}+1}$, giving two representations S_+, S_- (why are they nonisomorphic?).